

Progress Report Sprint 2

- Course Name: CS461 Senior Software Engineering Project I
- Date: 10/23/25
- Team Name: TaxonBodyMassML
- Team ID: CS.032
- Student Names and Emails:
 - Grant Pasquantonio (pasquang@oregonstate.edu)
 - Hailey Prater (praterh@oregonstate.edu)
 - Anish Chowdary Pendurti (pendurta@oregonstate.edu)

Summary

- Completed requirements elicitation
- Started prototyping web interface design
- Completed a functional data visualization and partitioning script
- Started combination of data file and taxonomy.

Progress vs Plan

PLANNED ITEM	STATUS (% OR DONE/PARTIAL/NOT DONE)	LINKS (ISSUE/PR)	NOTES/BLOCKERS	PLAN CHANGE (IF ANY)
DATA PREP	Partial	https://github.com/marknovak/TaxonBodyMassML/tree/data-partitioning/data_partition	May consider an alternative approach using normalization.	N/A
WEB DESIGN	Partial	https://github.com/marknovak/TaxonBodyMassML/tree/web-dev/web_dev	Will continue to improve layout/design in the coming weeks.	N/A
DATA COMBO	Partial	https://github.com/marknovak/TaxonBodyMassML/tree/data-combination	Need to get taxonomy data file from the professor	N/A
REQUIREMENTS	Done	https://github.com/marknovak/TaxonBodyMassML/tree/main/documentation	The list of requirements is added in the documentation folder in the GitHub repo.	N/A

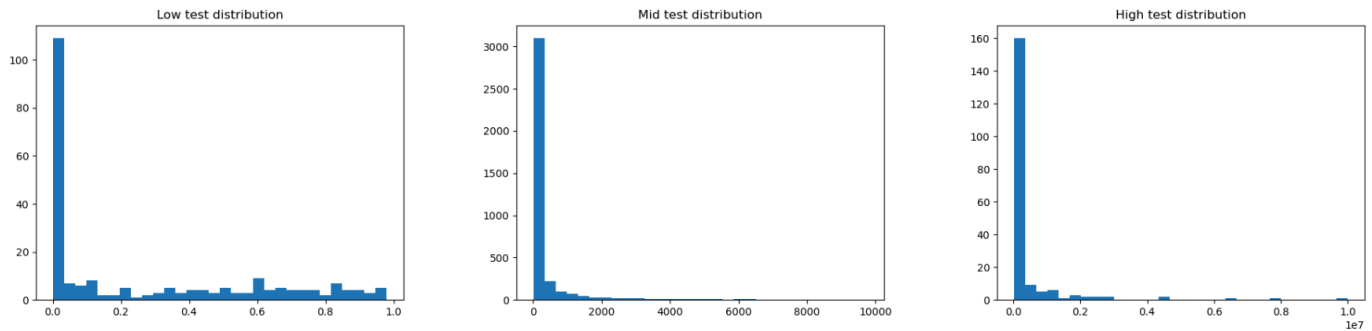
Evidence of Working Software

Data visualization latest update commit hash: c6178718aa3d4447f958cf4a441708f881398ff7

Web Design latest update commit hash: bd2bbd8b14ab96be12851cd290f7e3e16ba2b1c9

Link to UI: https://praterh.github.io/HaileysTaxonBodyMassML/web_dev/index.html

Design Plan Data Visualization



Risk and Quality

- As apparent in the data visualization, the distribution of mass is heavily skewed. The risk here is that our random sampling of test data may not be representative if we use a simple random sample. This means we may need to apply a data normalization method.
 - Owner: Grant
 - Status: in progress
 - Concrete actions: Apply logarithmic normalization to the mass variable as a new column of the data. Use this normalized variable for the test/train data split and potentially even the model itself so that we can use predictive models based on priors of a normal-like distribution.
- When merging the body mass data set and the catalog of life data, the outputted data file might not match the taxonomy and the species mass accurately. The risk here is that a incorrect mass might be matched to the incorrect species ruining the data.
 - Owner: Anish
 - Status: In-progress
 - Concrete Actions: To ensure the correct matching of species with their body mass data, preprocess the species names in both datasets to remove extra spaces and standardize them to lowercase. Use fuzzy matching with the fuzzywuzzy library to match species names that may differ slightly between datasets and only accept matches with a high confidence score (e.g., 90%). Finally, merge the datasets based on the best-matched species names and save the resulting data, ensuring the mass is correctly assigned to each species.

Next Goals

- Further the design and layout of the UI; this will work toward REQ-006, meeting accessibility standards
- Work on writing our predictive algorithm; this will allow for progress toward REQ-001 & REQ-002
- Continue optimizing the training/test data split; this falls under REQ-008 because it is necessary in having effective accuracy determination.
- Continue to merge our datasets; this shows our progress in REQ-007 because both of these sets are publicly accessible

Individual Contributions

Grant:

- Working on branch: <https://github.com/marknovak/TaxonBodyMassML/tree/data-partitioning>
- Created data visualization using pyplot to get an idea of the distribution based on the "mass_g" variable
- Created data partitioning script to split data into training and testing datasets
- Added labels to the plots for enhanced clarity.

Hailey:

- Working on branch: <https://github.com/marknovak/TaxonBodyMassML/tree/web-dev>
- Drew out a first draft of UI
- Created basic web development files for UI
- Incorporated histograms into the UI on the 'Data Visualization' page
- Made website publicly accessible (through [this link](#))

Anish:

- Working on branch: <https://github.com/marknovak/TaxonBodyMassML/tree/data-combination>
- Worked on data combination using pandas to get a data set of combined taxonomy of a species and their mass.
- Incorporated Open Life API into the script to allow data set combination.